

Outcome Definition

Children were classified as **at risk** if their score was below the 17th percentile for each of the following outcomes:

- BSID Cognitive Composite Score (CCS)
- BSID Language Composite Score (LCS)
- BSID Motor Composite Score (MCS)
- GSED

Predictors

We restricted all predictors to measurements obtained within the first year of life, as the goal is to predict later cognitive outcomes from early-life markers. The following categories of predictors were considered:

- **Demographics & Perinatal:** sex, maternal education, maternal age, gestational age
- **Anthropometry:** birth weight, birth length, weight-for-age z-score, height-for-age z-score, weight-for-height z-score
- **EEG Features:** Low Alpha, High Alpha, Beta, Gamma, Theta Beta Ratio, High Alpha Delta Ratio, Alpha Peak
- **MiniMorph Brain Volumes:**
 - Caudate, Putamen, Thalamus, Ventricles, Corpus Callosum, Supratentorial CSF, Supratentorial Tissue, Cerebellum
- **LoMINA-SC Brain Volumes:**
 - CSF, Gray Matter, White Matter, Corpus Callosum, Caudate, Lentiform, Hippocampus, Thalamus
- Age at scan

Brain Volume Adjustment

To account for head/brain size, we used **residual volumes** (fit a cubic regression on head circumference MRI and used standardized residuals).

Data Imputation

- Allowed to be imputed: demographics, perinatal variables, anthropometry.
- Not imputed: imaging features, EEG features, and outcomes (to preserve biological validity). Missing participants for these variables were excluded.
- Imputation was performed using **missForest** (non-parametric random forest-based imputation).

Sample Demographics by Site

For each site, we report both the **overall available sample size** for each outcome and the **effective analytic sample size** after restricting to participants with complete predictor sets. Sample sizes are listed in the order BSID CCS, BSID LCS, BSID MCS, and GSED, except for MINE, PRISMA-Zambia, and PRISMA-Kenya, where only GSED was available. For modality-specific analyses (e.g., EEG, MRI, EEG+MRI), participants missing that modality were excluded. Here, MRI refers to the union of MiniMorph and LoMINA-SC predictors. Baseline covariates, including demographics, perinatal variables, and anthropometry, were always included as predictors for all cohorts, regardless of the modality under consideration.

- UCT (Khula + SPACE)
 - Overall: 75, 68, 58, 199
 - EEG only: 46, 44, 34, 101
 - MRI only: 41, 37, 30, 127
 - MRI + EEG: 33, 32, 24, 79
- PRIMES
 - Overall: 68, 68, 68, 57
 - MiniMorph: 58, 58, 58, 48
- MINE
 - Overall: 148 (GSED)
 - MRI: 30 (GSED)
- PRISMA-Zambia
 - Overall: 133
 - MRI: 97
- PRISMA-Kenya
 - Overall: 93
 - MiniMorph: 47

Methodology

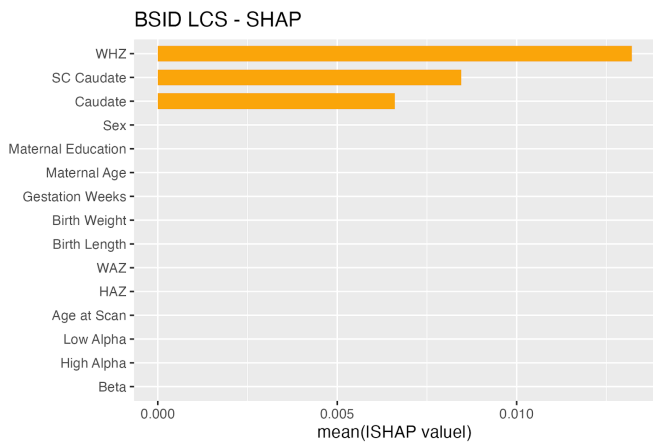
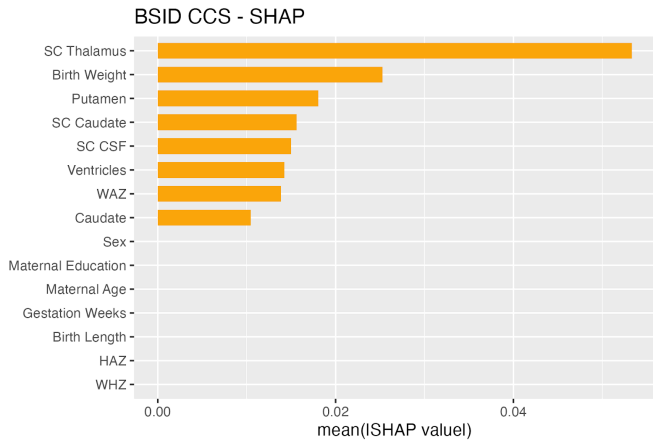
- **Algorithms:** XGBoost
- **Evaluation:** Five-fold cross-validation, reporting **recall**, **precision**, and **F1** for the risk class.
- **Feature Importance:** Global SHAP values were used to identify the most influential predictors across the dataset.

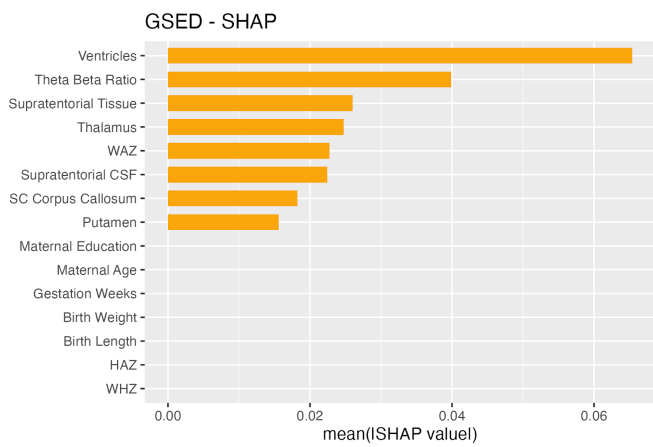
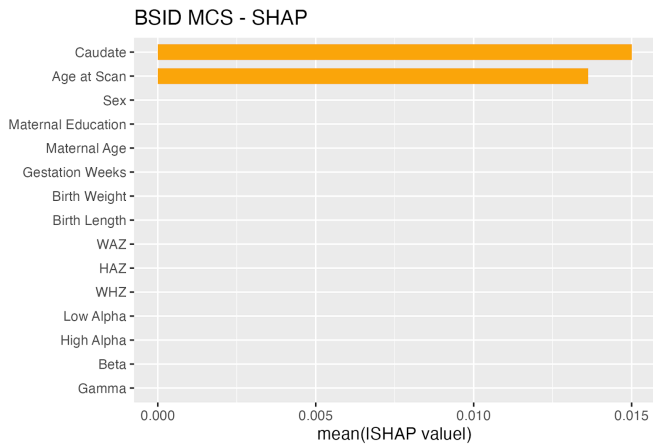
Results

XGBoost Performance

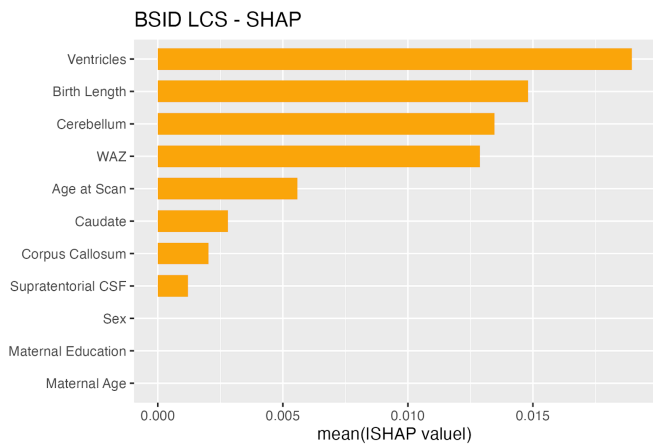
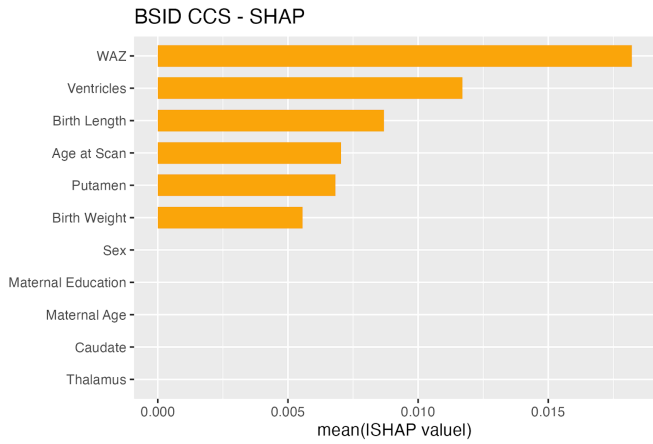
SHAP

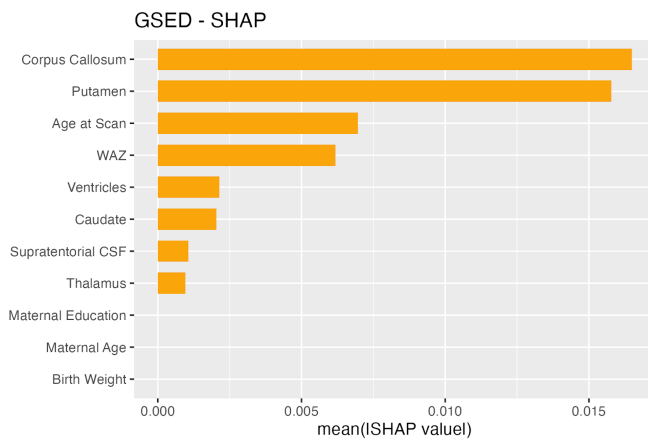
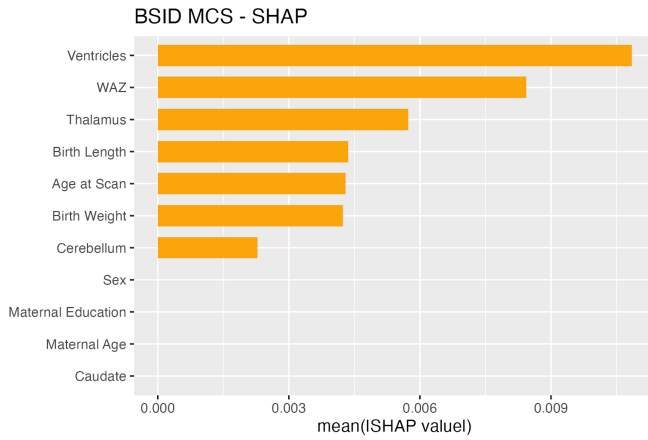
- UCT (Khula + SPACE) using MRI + EEG



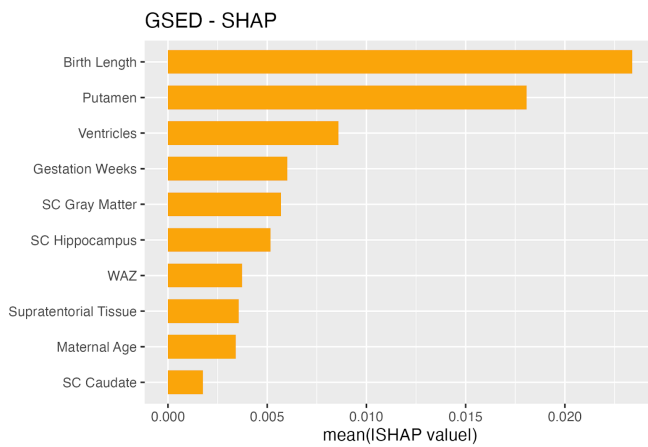


- PRIMES using MiniMorph

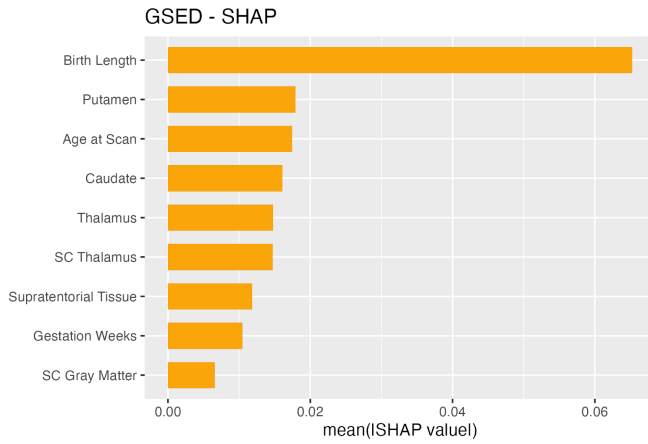




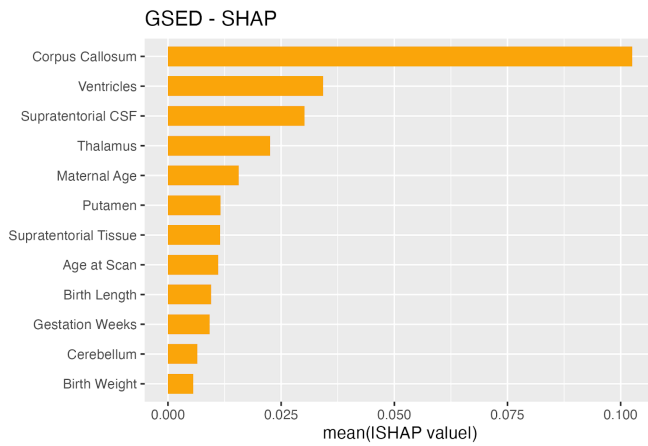
- MINE using MRI



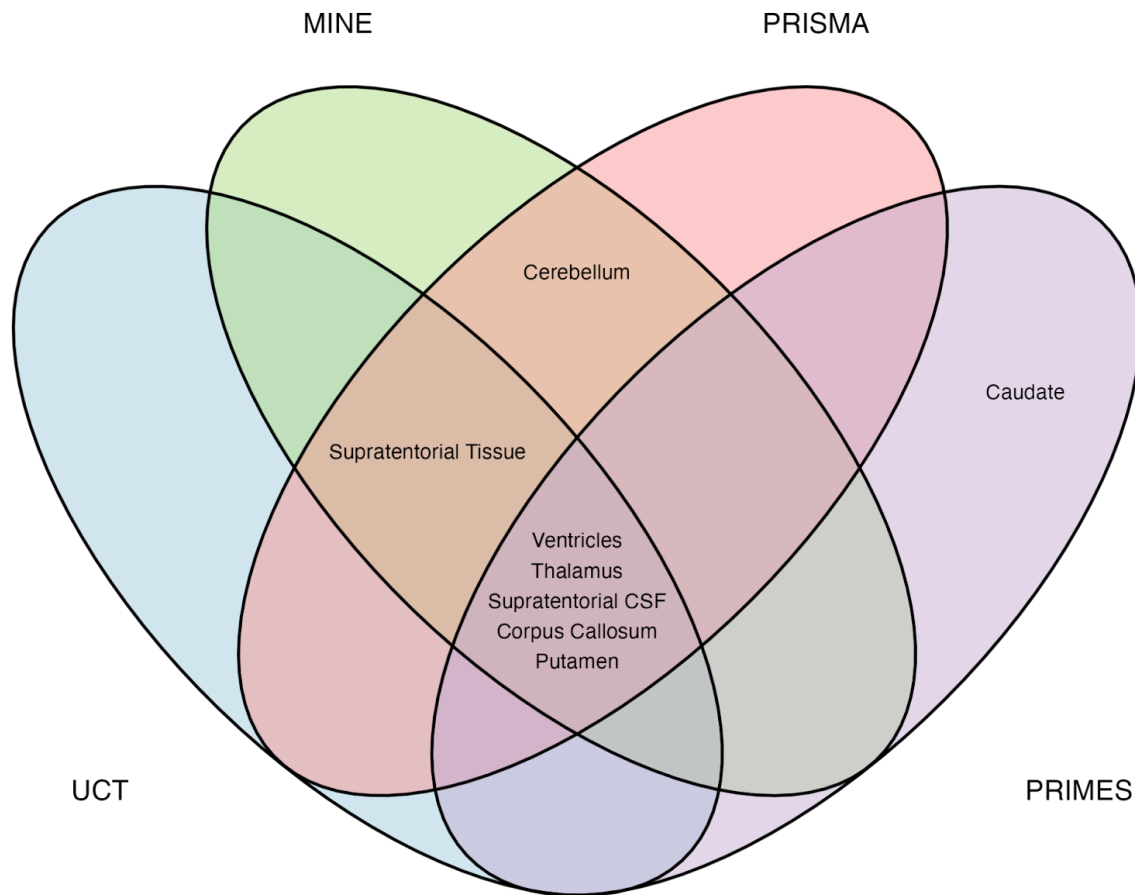
- PRISMA-Zambia using MRI



- PRISMA-Kenya using MiniMorph



Venn Diagram for GSED



Cross-Cohort Consistency and Performance

The Venn diagram highlights a notable degree of **consistency in SHAP-based feature importance across cohorts**. Several brain regions (e.g., ventricles, thalamus, supratentorial CSF, corpus callosum, and putamen) consistently emerged as influential predictors in multiple sites. This overlap suggests that despite cohort heterogeneity, the model identifies a stable set of neuroimaging features associated with developmental outcomes.

In terms of **classification performance**, MRI-based predictors were available across all cohorts, while UCT additionally provided EEG. At UCT, the **combination of MRI and EEG yielded the most balanced performance**, with higher F1 scores reflecting the complementary

strengths of structural imaging and electrophysiological data. Across the other cohorts, where only MRI was available, predictive accuracy was still reasonably good, especially for BSID.

The results clearly show that where available the combination of EEG and MRI leads to the overall best detection results, especially for the BSID scores (where available). This also holds true for GSED, which is generally harder to predict since it is measured at a much earlier age. Notably, the confusion matrices show very good detection performance of at-risk children across all cohorts and scores when utilizing MRI or MRI + EEG (the latter for UCT).

Brain-for-Age Percentiles: Methods and Plots

Overview

We constructed **brain-for-age percentile curves** from MiniMorph brain volumes to visualize developmental patterns across cohorts. Unlike anthropometric growth charts, where individual trajectories are often emphasized, here the focus was on estimating smooth age-dependent distributions of brain volumes. The methodology was developed in Chen et al. (2021) NeuroImaging.

Data Inputs

- **Age at scan**
- **MiniMorph brain volumes**, including Ventricles, Thalamus, Supratentorial CSF, Corpus Callosum, Putamen

Methodology

1. Age Binning and Empirical Distributions

- The sample was divided into bins according to **age at scan**.
- Within each age bin, we computed the **empirical distribution of brain volumes** for each region.

2. Distributional Regression (Local Fréchet Regression)

- We applied **local Fréchet regression** to the set of empirical distributions, using the **bin midpoints as inputs** and the **distributions as outputs**.
- This approach smooths the sequence of distributions across age, producing continuous age-dependent distribution functions.

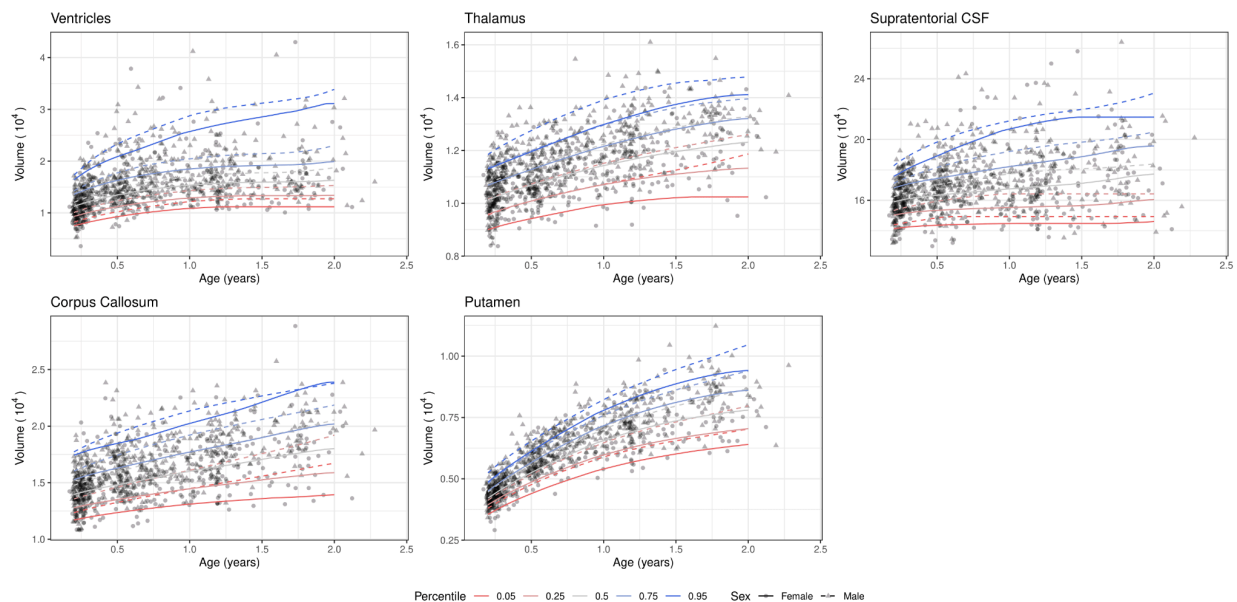
3. Percentile Curves

- From the smoothed age-dependent distributions, we extracted **percentile curves** (5th, 25th, 50th, 75th, and 95th).

- These curves provide a compact summary of the distributional changes in brain volumes with age.

Results and Interpretation

The use of **distributional regression via local Fréchet regression** allowed us to estimate smooth age-dependent percentiles from raw MiniMorph brain volumes. These plots serve as a descriptive complement to our predictive models, offering an interpretable framework for comparing structural brain development across cohorts.



- Chen, Yaqing, Paromita Dubey, Hans-Georg Müller, Muriel Bruchhage, Jane-Ling Wang, and Sean Deoni. "Modeling sparse longitudinal data in early neurodevelopment." *NeuroImage* 237 (2021): 118079.